

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	XXXXXXXXXX
Project Title	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the credit card approval process using machine learning techniques. By analyzing applicant information and historical credit records, the system predicts whether a credit card application should be approved or rejected. The solution reduces manual effort, improves decision accuracy, minimizes financial risk, and provides faster approval decisions through a user-friendly Flask web application.

Project Overview	
Objective	Develop a machine learning-based system that predicts credit card approval using applicant demographics and credit history, enabling faster and more accurate decision-making.
Scope	The project covers data preprocessing, feature engineering, model training, performance evaluation, and deployment of the best-performing model through a Flask web application for real-time prediction
Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manual verification is slow, time-consuming, and prone to human error, making it difficult to process applications efficiently and identify high-risk applicants accurately.

Impact	Delayed approvals, increased operational costs, inconsistent decisions, and higher financial risk due to manual processing and inaccurate applicant evaluation.
Proposed Solution	
Approach	Collect applicant and credit history data, preprocess the datasets, perform feature engineering, train multiple machine learning models (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting/XGBoost), select the best-performing model, and integrate it into a Flask web application for instant approval prediction.
Key Features	• Automated credit card approval prediction. • Data preprocessing and feature engineering. • Comparison of multiple ML algorithms. • Real-time prediction through a Flask web application. • Fast, accurate, and consistent decision support for financial institutions.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	System for data preprocessing, model training, and Flask deployment	Intel Core i5/i7 (or equivalent), 8 GB RAM, 100 GB Storage
Memory	RAM required for preprocessing and model training	8 GB

Storage	Space for datasets, trained models, and project files	10–20 GB Free Disk Space
Software		
Frameworks	Machine learning and web application framework	Python, Flask
Libraries	Data processing, visualization, and machine learning	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Joblib/Pickle
Development Environment	IDE and package management	VS Code / PyCharm, Anaconda Navigator
Data		
Dataset	Applicant and credit history datasets	application_record.csv, credit_record.csv (Kaggle Dataset)
Model Storage	Store trained ML model	.pkl or .joblib file

Cloud / Deployment		
Deployment Platform	Host prediction service	IBM Watson Machine Learning / Flask Local Server
Version Control	Source code management	Git & GitHub